

Analysis - Lay (OUTLINE)

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1. Section 13 - Topology of the Reals

- (Defn) Neighborhood and Deleted Neighborhood
- (Defn) Interior point
- (Defn) Boundary point
- (Defn) Open and Closed Sets
- (Thm) Equivalent definitions of open and closed sets
- (Thm) Arbitrary unions of open sets are open
Proof idea: Every point in the union lies in some open set. Take a neighborhood that lies within that set. It also lies within the union.
- (Thm) Finite intersections of open sets are open
Proof idea: Let c be in the finite intersection. Then for each set there is a neighborhood of c contained within that set. Each has a radius. consider the neighborhood with the smallest radius. Then this lies in the intersection.
- (Thm) Arbitrary intersections of closed sets are closed.
Proof idea: Way 1 - by complements of previous theorem. Use Demorgan's law. Way 2 - by definition of boundary.
- (Thm) Finite unions of closed sets are closed.